Ryan Hazlett

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EDUCATION

University of Toledo **August 2020 - present**
Ph.D. Candidate, Physics
Thesis Advisor: Prof. Eli Visbal

University of Oklahoma **August 2016 - May 2020**
Bachelor of Science, Astrophysics
Research Advisor: Prof. Karen Leighly

RESEARCH INTERESTS

I am a theoretical astrophysicist and cosmologist primarily studying the formation of the first stars and galaxies and the origins of supermassive black holes. I use both semi-analytic methods and hydrodynamical cosmological simulations to make predictions for the next generation of observatories like the James Webb Space Telescope.

EXPERIENCE

Research Assistant **December 2020 - present**
University of Toledo, Advisor: Prof. Eli Visbal

Undergraduate Research Assistant **April 2018 - May 2020**
University of Oklahoma, Advisor: Prof. Karen Leighly

NSF REU student **May 2018 - July 2018**
University of Oklahoma, Advisor: Prof. Karen Leighly

FYRE program student **October 2016 - May 2017**
University of Oklahoma, Advisor: Prof. Mukremin Kilic

GRANTS AND AWARDS

Doreen and Lyman Spitzer Senior Graduate Fellow in Astrophysics, $32,500 (University of Toledo) 2026 - 2027
Physics & Astronomy Outstanding Department Service Award (University of Toledo) 2023 - 2024
Astronomy Seminar Best Graduate Student Presentation (University of Toledo) Spring 2021
Department of Physics & Astronomy Outstanding Senior Award (University of Oklahoma) 2020
Oklahoma State Regents Institutional Nominee Award, $38,000 (University of Oklahoma) 2016 - 2020

SERVICE

Astronomy Journal Club Host (University of Toledo) Fall 2023 - present
Physics & Astronomy Python Workshop Organizer (University of Toledo) Summer 2023 - present
Department of Physics & Astronomy Photographer (University of Toledo) Fall 2024 - Summer 2025
Prospective Student Host (University of Toledo) Fall 2024 - Spring 2025
Physics & Astronomy Grad Task Organizer (University of Toledo) Fall 2022 - Spring 2024

MENTORING

Merrill Edleman, University of Toledo Undergraduate March 2025 - present
Project: Calibrating to Reionization in the THESAN Simulations at high redshift

Anneysha Bahar, Ottawa Hills High School Student October 2023 - May 2024
Project: Calibrating a Semi-analytic Model to Renaissance Simulation Metal Bubbles

Presented poster at 2023-2024 Junior Science & Humanities Symposium

Thomas Behling, University of Toledo Undergraduate — May 2023 - May 2024
Project: Evaluating Reionization Prescriptions in Semi-analytic Models
Submitted first author paper to JCAP

TEACHING AND OUTREACH

Lab TA, *General Physics I* (University of Toledo) — Summer 2025 - Spring 2026
Lab TA, *Physics-Sci & Eng Majors I-B1* (University of Toledo) — Fall 2020 - Spring 2021
Lunar Sooners Member (University of Oklahoma) — 2016 - 2018

CONFERENCE PRESENTATIONS

- *AAS* 247th *Meeting*, Phoenix, AZ. January 2026. **Talk:** "From Population III Stars to Heavy Black Hole Seeds: Modeling Early Structure Formation with Semi-analytic Models and Simulations"
- *Cosmic Frontier Center Conference*, UT Austin, Austin, TX. May 2025. **Poster:** "High-Redshift Atomic Cooling Halos generally lead to Intermediate-Massive Black Hole Seeds"
- *First Light Conference*, MIT, Cambridge, MA. June 2023. **Poster:** "Calibrating Semi-analytic Models of the First Galaxies with Hydrodynamical Simulations to Enable Computationally Efficient Observational Predictions"
- *Artificial Intelligence and Maching Learning at OU*, Norman, OK. September 2019. **Poster:** "Quantifying Massive Outflows using the MCMC Spectral Synthesis Code SimBAL"
- *AAS* 233rd *Meeting*, Seattle, WA. January 2019. **Poster:** "Massive Outflows in CII Low-Ionization Broad Absorption Line Quasars"

PUBLICATIONS

1. Visbal, E., Bryan, G. L., **Hazlett, R.** *Massive Population III Galaxies Near a GN-z11 Analog: The Role of the Baryon-Dark Matter Streaming Velocity*, submitted to JCAP, 2026, arXiv:2609.35941
2. **Hazlett, R.**, Visbal, E., Bryan, G. L., Kulkarni, M. *Abundant Heavy Black Hole Seeds from Moderate Lyman-Werner Radiation*, submitted to JCAP, 2026, arXiv:2608.13656
3. Feathers, C. R., Visbal, E., Murray, S., **Hazlett, R.**, Ma, Y. *Exploring the High-Redshift 21-cm Signal via Self-Consistent Simulations using Artificial Neural Network Emulation*, accepted by MNRAS, 2026, arXiv:2605.29876
4. **Hazlett, R.**, Mead, J., Visbal, E., Bryan, G. L., Mac Low, M.-M., Kulkarni, M., Andersson, E. P., Brauer, K., Wise, J. H. *From Primordial Stars to Early Galaxies: A Semianalytic Model Calibrated with AEOS and Renaissance*, ApJ, 1009, 3, 2026
5. Visbal, E., **Hazlett, R.**, Bryan, G. L. *LAP1-B is the First Observed System Consistent with Theoretical Predictions for Population III Stars*, ApJ, 993, L17, 2025
6. Behling, T., **Hazlett, R.**, Kulkarni, M., Visbal, E. *Evaluating the Accuracy of Reionization Prescriptions in Semi-analytic Models of the First Stars and Galaxies*, submitted to JCAP, 2025, arXiv:2508.04808
7. **Hazlett, R.**, Kulkarni, M., Visbal, E., Wise, J. H. *A Framework to Calibrate a Semianalytic Model of the First Stars and Galaxies to the Renaissance Simulations*, ApJ, 978, 13, 2025
8. Feathers, C. R., Kulkarni, M., Visbal, E., **Hazlett, R.** *A Global Semianalytic Model of the First Stars and Galaxies Including Dark Matter Halo Merger Histories*, ApJ, 962, 62, 2024